Seasonal Chlorophyll-a Hotspots Along the Southwestern South Atlantic Margin

Literature review: in progress
Public portfolio excerpt from the revised detailed outline and annotated bibliography dated October 2, 2026. This excerpt is not a completed research paper or literature synthesis.

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Context: GEOG 434 Hydrology and Environment; related oceanographic visualization work with Dr. Slowey.

Research question and fixed class scope

Where do chlorophyll-a hotspots recur along the southwestern South Atlantic margin between 30 degrees S and 55 degrees S during 2018-2024, and how do their seasonal patterns relate to sea-surface-temperature gradients and bathymetric setting? The class domain is 70-45 degrees W and 55-30 degrees S. Related Slowey visualizations extend into 2026; those longer records remain distinct from the class-paper period.

What the literature needs to explain

Seasonal timing. Compare when chlorophyll maxima recur across the Rio de la Plata, Argentine and Patagonian shelves and waters north of the Falkland Islands, rather than assuming one seasonal pattern for the entire margin.

Physical setting. Compare shelf-break, tidal, estuarine and boundary-current fronts. Bathymetry, wind, freshwater influence and mixing can support different mechanisms in different sectors.

Biological context and uncertainty. Examine nutrient supply, phytoplankton responses and ocean-color retrieval limits. Whale habitat motivates the project, but the planned class analysis contains no whale-observation layer or direct prey measurements.

Current writing state

Eleven regional papers are annotated and cited in the working outline. Full-text evidence extraction, a source-comparison matrix and a thematic synthesis remain review tasks. The current material provides questions and expectations for analysis; it does not establish hotspot persistence, a trend or a causal explanation.

From literature to the planned analysis

Planned methods, not completed findings

The working outline proposes monthly and austral-seasonal summaries of chlorophyll and temperature, temperature-gradient/front comparisons, hotspot persistence and comparisons with bathymetric setting. The final temperature source, named bathymetric release, common analysis grid, valid-coverage rules and several aggregation choices still need to be documented before stronger statistical interpretation.

Sector comparisons would be descriptive associations. Shared seasonality, temporal dependence, different spatial support and uncertainty in the source products must remain part of interpretation.

Completed preparation that supports the next stage

The retained project includes daily chlorophyll acquisition/conversion and class visualizations for 2018-2024, longer Slowey chlorophyll and currents map sequences, and requested 2019 comparison layouts covering 22 selected dates. These products demonstrate reproducible preparation and visualization. Seasonal statistics and the final paper remain work in progress.

Product differences matter

NOAA CoastWatch chlorophyll: a VIIRS + OLCI DINEOF reconstructed product.
Copernicus GLORYS12V1: modeled ocean reanalysis, including near-surface currents and temperature.
NASA JPL MUR v4.1: merged/interpolated foundation sea-surface-temperature analysis.

Calendar-day pairing does not make these simultaneous observations. Grid spacing does not establish effective observing resolution. GLORYS assimilates sea-surface temperature, so a MUR-versus-GLORYS comparison is not an independent validation.

Two examples from the annotated literature

Acha, E. M., Mianzan, H. W., Guerrero, R. A., Favero, M., & Bava, J. (2004). Marine fronts at the continental shelves of austral South America: Physical and ecological processes. *Journal of Marine Systems, 44*(1-2), 83-105. doi:10.1016/j.jmarsys.2003.09.005

Carranza, M. M., Gille, S. T., Piola, A. R., Charo, M., & Romero, S. I. (2017). Wind modulation of upwelling at the shelf-break front off Patagonia: Observational evidence. *Journal of Geophysical Research: Oceans, 122*(3), 2401-2421. doi:10.1002/2016JC012059

Excerpt prepared with Codex under Ethan DeMelo's oversight from the existing working outline and October 19 literature handoff. The complete working document and internal team planning remain in the local research folder.